

KANEHOE MCAS, HI, USA

WMO: 911760

Lat: 21.450N

Lon: 157.768W

Elev: 7

StdP: 101.24

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22519

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.7	18.8	11.7	8.6	21.4	12.6	9.1	21.7	10.6	22.4	9.3	22.1	0.9	180	0.341

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	4.3	29.9	24.0	29.2	23.7	28.8	23.6	25.4	27.9	24.9	27.6	24.5	27.4	4.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	19.6	26.9	24.0	18.9	26.7	23.5	18.3	26.5	77.4	28.3	75.5	27.5	73.7	27.4	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.5	7.0	15.0	31.0	3.5	1.6	12.5	32.1	10.4	33.0	8.5	33.9	5.9	35.1		
			13.2	26.5	1.4	1.3	12.2	27.4	11.4	28.2	10.6	28.9	9.6	29.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.8	23.0	22.9	23.2	23.9	24.7	25.6	26.0	26.5	26.6	26.1	24.9	23.7	(6)
(7)		DBStd	1.73	1.34	1.26	1.24	1.15	1.14	0.81	1.01	0.89	0.88	1.09	1.10	1.17	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5388	402	361	409	416	456	468	495	512	498	499	447	425	(10)
(11)		CDD18.3	2346	143	127	151	166	198	218	237	253	248	241	197	167	(11)
(12)		CDH23.3	14483	433	332	439	612	1066	1531	1857	2244	2212	1972	1165	620	(12)
(13)	CDH26.7	1609	11	8	16	23	69	148	240	371	377	253	78	16	(13)	
(14)	Wind	WSAvg	3.7	3.3	3.6	3.8	4.0	3.6	4.0	4.2	4.0	3.6	3.5	3.6	3.5	(14)
(15)	Precipitation	PrecAvg	828	131	74	79	93	54	30	45	38	43	72	92	108	(15)
(16)		PrecMax	1708	354	276	236	565	179	99	101	94	115	259	360	400	(16)
(17)		PrecMin	326	10	4	5	15	1	10	13	11	7	18	18	11	(17)
(18)		PrecStd	291	99	60	65	106	49	19	25	22	25	59	77	88	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.8	27.7	27.9	28.2	29.0	29.1	29.6	30.5	30.9	30.2	29.1	27.9	(19)
(20)			MCWB	21.5	21.6	21.8	22.5	23.3	23.4	23.9	24.3	24.6	24.1	23.5	22.3	(20)
(21)		2%	DB	26.9	26.6	27.1	27.4	27.9	28.6	29.0	29.6	29.9	29.1	28.1	27.2	(21)
(22)			MCWB	21.3	21.1	21.3	21.8	22.7	23.1	23.7	23.9	24.2	23.8	23.0	22.0	(22)
(23)		5%	DB	26.2	26.0	26.3	26.6	27.4	28.0	28.5	29.0	29.1	28.6	27.5	26.4	(23)
(24)			MCWB	21.2	20.9	21.1	21.6	22.3	22.8	23.5	23.7	23.9	23.7	22.7	21.6	(24)
(25)		10%	DB	25.5	25.2	25.6	26.1	27.0	27.5	27.9	28.4	28.5	27.9	27.0	26.0	(25)
(26)			MCWB	20.9	20.7	21.0	21.4	22.1	22.6	23.2	23.5	23.6	23.3	22.4	21.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.0	23.5	23.5	23.8	24.2	24.6	25.7	25.8	26.2	25.7	25.0	23.9	(27)
(28)			MADB	25.5	25.5	26.0	26.3	27.2	27.5	27.5	28.1	28.7	28.1	27.5	25.9	(28)
(29)		2%	WB	23.0	22.7	22.8	23.0	23.7	24.1	25.0	25.2	25.2	24.8	24.1	23.2	(29)
(30)			MADB	24.9	25.0	25.3	25.7	26.7	27.1	27.2	27.9	28.2	27.6	26.8	25.6	(30)
(31)		5%	WB	22.4	21.9	22.2	22.5	23.2	23.6	24.4	24.6	24.7	24.3	23.6	22.8	(31)
(32)			MADB	24.6	24.5	24.8	25.4	26.2	26.8	27.1	27.6	27.8	27.4	26.4	25.3	(32)
(33)		10%	WB	21.8	21.4	21.7	22.0	22.8	23.2	24.0	24.2	24.2	23.9	23.1	22.3	(33)
(34)			MADB	24.3	24.2	24.4	25.0	25.9	26.5	26.9	27.3	27.5	27.1	25.9	24.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	4.7	4.6	4.2	4.4	4.2	4.3	4.3	4.2	4.2	4.5	(35)	
(36)			MCDBR	5.9	5.8	5.6	4.9	4.8	4.5	4.7	4.9	5.0	4.8	4.9	5.1	(36)
(37)			MCWBR	3.3	3.2	2.9	2.6	2.4	2.3	2.4	2.4	2.3	2.4	2.7	2.9	(37)
(38)		5% WB	MCDBR	5.3	4.9	5.1	4.5	4.5	4.2	4.2	4.4	4.5	4.6	4.7	4.7	(38)
(39)			MCWBR	3.6	3.2	3.1	2.7	2.6	2.4	2.5	2.5	2.5	2.6	2.9	3.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.365	0.373	0.381	0.391	0.383	0.370	0.369	0.368	0.361	0.379	0.378	0.360	(40)	
(41)		taud	2.567	2.533	2.521	2.480	2.532	2.575	2.589	2.593	2.616	2.568	2.562	2.602	(41)	
(42)		Ebn at Noon	909	921	924	913	910	914	915	922	930	903	888	902	(42)	
(43)		Edh at Noon	93	101	107	113	106	101	99	99	96	97	93	87	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.37	4.93	5.66	6.39	6.87	7.20	7.07	6.78	6.34	5.32	4.48	4.04	(44)	
(45)		RadStd	0.18	0.35	0.59	0.36	0.38	0.20	0.15	0.27	0.19	0.16	0.11	0.29	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.37	N/A	N/A	+0.60	+0.39	+0.45	N/A	N/A	+139	+140	(46)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+68	+68	(47)

Nomenclature: See separate page